Qing Li | Curriculum Vitae

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RESEARCH INTEREST

Robust and Continual Reinforcement Learning, Policy Optimization under Non-Stationary Dynamics, Reward Dynamics and Ergodicity, Long-Horizon Decision-Making, Multi-Agent Reinforcement Learning.

EXPERIENCE

Xinjiang University Urumqi, China

Student Researcher - Generative Models and Multi-Agent Reinforcement Learning *07/2025 - Present*

- **Generative Policy Optimization:** Developed a diffusion model-enhanced multi-agent proximal policy optimization (DM-MAPPO) framework by integrating conditional reverse denoising into policy learning for cooperative decision-making under dynamic environmental conditions.
- **Multi-Agent Decision-Making:** Formulated cooperative odor source localization as a decentralized partially observable Markov decision process (Dec-POMDP) with unknown transition probabilities, using centralized training and decentralized execution (CTDE).
- **Reward Design and Evaluation:** Designed a composite reward function balancing exploration, exploitation, collision avoidance, and search efficiency; evaluated the proposed approach against baseline algorithms in CFD-based simulation environments.

Student Researcher - Deep Reinforcement Learning *09/2024 - 06/2025*

- **Non-Stationary Environment Modelling:** Constructed time-varying simulation datasets using Python and computational fluid dynamics (CFD) to investigate autonomous decision-making under changing environmental conditions.
- **Event-Driven Decision Modelling:** Developed event-driven control models and priority-based decision mechanisms to represent asynchronous state transitions and coordinate system behaviour.
- **Temporal Feature Modelling:** Implemented and tuned Transformer-based neural network architectures in PyTorch for autonomous source localization, focusing on the use of environmental observations in sequential decision-making.

North University of China Tai Yuan, China

Undergraduate Research Assistant - Intelligent Device for Autonomous Rubik's Cube Solving *09/2018 - 06/2022*

- **Control System Implementation:** Designed, assembled, and debugged a pneumatic actuation-based control system for automated Rubik's Cube manipulation, integrating mechanical actuation and sequential control operations.

PROJECTS

- **Event-Driven Reinforcement Learning with Transformer-Based Feature Evaluation:** Developed an event-driven learning framework combining Transformer-based feature evaluation, Enhanced Group Relative Policy Optimization (EGRPO), and Prioritized Colored Petri Nets (PCPN) to model asynchronous decision processes and priority-dependent state transitions. Prepared a research manuscript describing the proposed framework and its application to event-driven decision-making, and submitted it to Applied Soft Computing Journal.
- **Generative AI-Enhanced Multiagent DRL:** Propose a Diffusion Model-Enhanced Multiagent Proximal Policy Optimization (DM-MAPPO) for cooperative odor source localization. Implement the decentralized partially observable Markov decision process (Dec-POMDP) framework using PyTorch. I have submitted it to Neurocomputing Journal.

EDUCATION

Xinjiang University Urumqi, China

M.Res. in Control Science and Engineering; GPA: 3.59/4.00 *09/2024 – 06/2027 (Expected)*

- Selected Coursework: Machine Learning and Applications, Optimization Theory and Methods, Matrix Analysis.

North University of China Taiyuan, China

B.Eng. in Electrical Engineering and Intelligent Control; GPA: 3.32/4.00 *09/2018 – 06/2022*

- Selected Coursework: Advanced Mathematics, Linear Algebra, Complex Variables Functions & Integral Transforms, Probability Theory & Mathematical Statistics.

PUBLICATIONS

[1] **Qing Li**, Jie Yuan*, Li Zhu, Ningning Zhang, Erkenbieke Jia. "Autonomous odor source localization using event-driven deep reinforcement learning and transformer-based feature evaluation." *Applied Soft Computing*, 2026. Status: Under Review. [JCR Q1, IF: 7.8]

[2] **Qing Li**, Jie Yuan*, Ningning Zhang, Erkenbieke Jia. "Generative AI-enhanced multiagent deep reinforcement learning for cooperative odor source localization." *Neurocomputing*, 2026. Status: Under Review. [JCR Q1, IF: 6.7]

SKILLS

Technical Skills: Proficient in Python, CFD ANSYS, C/C++, MATLAB, Origin, Visio, LaTeX, Office.

Languages: Chinese (native); English (IELTS 6.5).

HONORS & AWARDS

- Academic Scholarship 2024, 2025
- National Encouragement Scholarship (Top 3%) 2019, 2020

REFERENCES

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